(217) 721-8373
lijiaxi2018@gmail.com

Jiaxi Li

PhD Candidate

LinkedIn: lijiaxi2018
GitHub: lijiaxi2018

Research Interests

My research focuses on designing and optimizing large language model (LLM) systems for improved performance through adaptive resource scheduling and hardware-aware optimization. My recent projects include energy-efficient prefill-decode disaggregated LLM serving and SLO-aware serving for multi-agent LLM systems.

Education

University of Illinois Urbana-Champaign Urbana, IL

Ph.D. in Computer Science, 4.00/4.00 2023/08 – 2028/05 (Expected)

- Advisor: Klara Nahrstedt

M.S. in Computer Science (thesis), 4.00/4.00 2022/08 – 2023/05

- Advisors: Klara Nahrstedt and Lawrence Angrave

B.S. in Computer Science (thesis), 3.91/4.00 2018/08 – 2022/05

Experience

IBM Research 2025/05 – 2025/08

Research Extern, Mentors: Eun Kyung Lee and Yue Zhu Yorktown Heights, NY

- **Project: Energy-Efficient LLM Inference Clusters via Prefill and Decode Disaggregation**
- Designed an SLO-aware energy-efficient request scheduler for disaggregated LLM inference clusters.
- Benchmarked performance and energy consumption of disaggregated LLM serving using vLLM and LMCache across multiple KV cache transfer paths (GPU peer-to-peer, CPU offloading, disk offloading).
- Proposed a novel serving-efficiency metric for prefill and decode stages, and modeled the efficiency–GPU frequency tradeoff to identify additional energy-saving opportunities from dynamic frequency scaling.
- **Published one workshop paper at EuroMLSys 2026 [C1].**

Selected Projects

SLO-Optimized Serving for Multi-Agent LLM Systems 2026/02 – Present

- Designed a latency- and quality-aware inference scheduler for Kubernetes-based multi-agent LLM systems, which supports heterogeneous GPUs and LLM/VLM deployments.
- Explored semantics-aware adaptive workflow planning and subtask routing for multi-agent execution.
- Proposed content-aware KV cache management via adaptive eviction and storage tier promotion/demotion.

360-Degree Video Analytics for Situation Awareness in Firefighting 2022/08 – 2024/08

- Built ST-360, a real-time 360-degree video analytics system that improves firefighter safety and productivity by detecting important objects and events from helmet-mounted cameras.
- Devised a spatiotemporal frame filtering algorithm for 360-degree video streaming and analytics, achieving a 50% reduction in end-to-end analytics latency compared to state-of-the-art methods.
- Realized a distortion-free object detection framework for 360-degree videos, outperforming existing approaches with a 19% improvement in F1 score and a 25% reduction in GPU memory usage.

- **Published one workshop paper at NOSSDAV 2023 [C2] and one journal paper at ACM TOMM [J1].**

ClassTranscribe: Accessible Online Learning Platform 2021/01 – 2023/08

- Architected the backend for ClassTranscribe, an online lecture platform with over 6,000 active users.
- Wrote algorithms to analyze image and text content from lecture videos, and created multimodal accessibility features to enhance learning experiences for students with hearing and vision impairments.
- **Published two conference papers at ASEE [C3–C4] and one thesis [T1].**

Publications

Journal

- [J1] **Jiaxi Li,** Jingwei Liao, Bo Chen, Anh Nguyen, Aditi Tiwari, Qian Zhou, Zhisheng Yan, Klara Nahrstedt. "ST360: Spatial–Temporal Filtering-Based Low-Latency 360-Degree Video Analytics Framework." **ACM TOMM,** 2024.

Conference & Workshop

- [C1] **Jiaxi Li,** Yue Zhu, Bo Chen, Eun Kyung Lee, Klara Nahrstedt. "Revisiting Disaggregated Large Language Model Serving for Performance and Energy Implications." **EuroMLSys**, 2026.
- [C2] **Jiaxi Li,** Jingwei Liao, Bo Chen, Anh Nguyen, Aditi Tiwari, Qian Zhou, Zhisheng Yan, Klara Nahrstedt. "Latency-Aware 360-Degree Video Analytics Framework for First Responders Situational Awareness." **NOSSDAV,** 2023.
- [C3] **Jiaxi Li,** Colin P. Lualdi, Yijun Lin, Aarya Bhatia, Jihong Cai, Sujit Varadhan, Rob Kooper, Lawrence Angrave. "The Inclusive Glossary: An Embedded, Interactive Approach to Accessible and Inclusive Learning." **ASEE,** 2023.
- [C4] Lawrence Angrave, **Jiaxi Li,** Ninghan Zhong. "Creating TikToks, Memes, Accessible Content, and Books from Engineering Videos? First Solve the Scene Detection Problem." **ASEE,** 2022.
  NEE Travel Grant Student Award

Thesis

- [T1] **Jiaxi Li.** "Image and Text Analytic Systems for Accessible Online Learning." **Master of Science Thesis,** 2023.

Academic Services

Reviewer, ACM TOMM, Computers & Education

Teaching Experience

Course Assistant, UIUC CS 361 Probability & Statistics for Computer Science 2019/09 – 2020/05
Course Assistant, UIUC CS 125 Introduction to Computer Science 2019/01 – 2019/05

Skills

Programming Languages:	C, C++, Python, Java, Go, Rust, JavaScript, SQL, HTML/CSS
Frameworks & Libraries:	vLLM, LMCache, scikit-learn, OpenCV, PyTorch, React, FastAPI, Spring Boot
Platforms:	Linux, Windows, Android, iOS, NVIDIA Jetson
Tools:	Docker, Kubernetes, Git, CI/CD, AWS, PostgreSQL, MongoDB
Languages:	English, Chinese, Japanese